

Vipin Kumar

AI Engineer

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Professional Experience

Lead Computer Vision Engineer

10/2022 – present

Add Innovations

- Led end-to-end deployment of a real-time AI monitoring system for **Denso**, processing feeds from **5 CCTV cameras** simultaneously; achieved **99.13% human detection accuracy at 10 FPS**, with rule-based violation detection and alarm triggering.
- Developed a **high-performance anomaly detection model** for **YKK tape fabric inspection**, achieving **99% accuracy** and outperforming **PaDiM and PatchCore** in both accuracy and inference speed through **model quantization**.
- Delivered an **automated eyewear inspection system** for **Lenskart**, combining **YOLOv8 + PaddleOCR** for curved and irregular text recognition, OpenCV-based symmetry checks (temples & rims), and deep learning-based dent detection.
- Built an **X-ray-based defect inspection pipeline** for **Munitions India Limited**, enabling multi-defect segmentation, defect size measurement, and OCR-based serial number extraction from noisy X-ray imagery.
- Implemented a **Tare Weight Detection system** for **Indian Oil**, designing a two-stage, multi-model OCR pipeline robust to **rotation, dirt, and damaged paint**, achieving **98.2% accuracy**.
- Developed an **AI-driven gear inspection solution** for **Sona Comstar**, achieving **98.3% accuracy across 8 categories** and reducing inspection cycle time to **10 seconds for 60 images** using **TensorRT acceleration**; collaborated with hardware and QA teams to meet real-time constraints.
- Designed a **deep learning-based defect segmentation and measurement system** for **YKK buttons**, processing **1,000+ images per minute** under variable lighting using batch inference and TensorRT.
- Implemented a **single-camera laser triangulation system** to measure screw head height with millimetre accuracy and classify screws as tight, loose, or over-tight for industrial quality inspection.
- Proficient in working with industrial cameras, Line-scan, area-scan, and CCTV (FLIR, Baumer, Hikvision, Dewinter); camera calibration for accurate measurement and object tracking.

Data science

04/2022 – 10/2022

Culinda Inc.

Performed data processing, preparation, and exploratory data analysis (EDA) on IoT sensor data, and trained LSTM autoencoders for anomaly detection on the Databricks platform. Conducted sensor pattern recognition, researched GAN architectures and LLMs, and developed a Text-to-SQL model.

Artificial intelligence (AI) Intern

01/2022 – 04/2022

BlobCity Inc.

Contributed to the open-source AutoAI (BlobCity) no-code AutoML framework for regression and classification. Designed automated data cleaning and preprocessing pipelines, enhanced the model search engine for optimal algorithm selection, implemented hyperparameter tuning.

Skills

Programming & Frameworks: Python, SQL, PyTorch, TensorFlow, Keras

Computer Vision & AI: YOLOv8, CNNs, Transformers, Object Detection, Segmentation, OCR, Anomaly Detection

Tools & Deployment: OpenCV, FastAPI, Flask, TensorRT, MLOps, CI/CD, ETL, Data Visualization

Hardware & Edge: Nvidia Jetson Orin Nano, FLIR, Baumer, Hikvision, Multi-camera setups

Projects

AgroSmart: Developed an AI-powered mobile app to optimize crop yields through soil analysis, disease detection, and fertilizer recommendations, improving production efficiency for farmers.

Education

- **B Tech (CSE)**, from Galgotias College of Engineering & Technology Greater Noida **CGPA: 7.87**

Achievements & Certificate

- Lead a team in SIH and got **1st rank in Smart India Hackathon 2022** for the problem Statement "Prepare AI readyDataset" under Ministry of Electronics and Information Technology.
- Runner up in an international **UNESCO India-Africa Hackathon**.
- **Won silver** medal and secured **65th Rank** in kaggle Ubiquant Market Prediction competition.
- **Achieved 5 bronze medals in Kaggle competitions** for Bengali.AI Speech Recognition, Linking Writing Processes to Writing Quality, and LLM - Detect AI Generated Text etc, **4x expert with under 400th rank Globally on kaggle**.
- Geoprocessing GIS using Python Certificate by Indian Institute of Remote Sensing
- Google Data Analytics Certificate by Coursera & Computer Vision with Opencv and Deep Learning by Jose Portilla
- Idea **evaluator at Smart India Hackathon 2024**, evaluated 100 ideas.
- Invited as a **judge in Galgotias International Hackathon**.
- Idea **evaluator at STATATHON Hackathon 2025**, evaluated 200 ideas.